

SECR3242 & SECR3941 Group Project

Course: SECR3242 - Internetworking Technologies

Semester: 2024-2025-02

Total Marks: **20%** (Report: 5%, Presentation: 5%, Packet Tracer: 10%)

Project Title: Enterprise Network Design & Implementation

Group Formation

- Form groups of **3 or 4 members**.
- Each group must **register their team and project topic by 20 June 2025**.

Project Objective

Design a **realistic network** for an **organization, university campus, or company** using **Cisco Packet Tracer**, implementing key networking concepts learned in class.

Scenario (Choose One)

1. **Corporate Office Network** (HQ + 2 Branch Offices)
2. **University Campus** (Main Campus + 2 Satellite Campuses)
3. **Hospital Network** (Main Hospital + 2 Clinics)
4. **E-Commerce Company** (Data Center + Office + Warehouse)

Network Requirements

Your design **must include**:

- **Minimum 3 routers** (more for better routing complexity).
- **Single Area OSPF** (All routers must be in **Area 0**).
- **5 ACLs (Standard/Extended)** for security & traffic control.
- **6 NAT Configurations** (Static, Dynamic, PAT).
- **VPN & IPsec** (Secure remote branch connectivity).
- **QoS Policies** (Prioritize VoIP/Video traffic).

Deliverables

1. Report (5%)

- **Cover Page** (Group members, ID, Project Title)
- **Introduction** (Scenario, Objectives)
- **Network Design** (Diagram + Explanation)
- **Configuration Details** (OSPF, ACLs, NAT, VPN, QoS)
- **Testing & Verification** (Screenshots of `show` commands)
- **Conclusion** (Challenges, Lessons Learned)
- **References** (If any)

Format: PDF (Max **10 pages**, Arial 11pt, 1.5 spacing)

2. Presentation (5%)

- **5-10 minute** presentation (PPT/Google Slides).
- **Explain:**
 - Network topology & design choices.
 - Key configurations (OSPF, ACLs, NAT, VPN, QoS).
 - Demo of Packet Tracer (live or recorded).
- **Q&A session** (2-3 mins).

3. Packet Tracer File (10%)

- Submit **.pka** file with:
 - Fully configured network.
 - Test cases (e.g., ACL blocking traffic, VPN working).
 - **Working** OSPF, NAT, VPN, ACL.

Evaluation Criteria

Component	Marks	Criteria
Report	5%	Clarity, completeness, technical accuracy
Presentation	5%	Delivery, content, engagement
Packet Tracer	10%	Functionality, correct configurations, complexity

Submission Deadline

- **Report & Packet Tracer: 5 July 2025** via Elearning
- **Presentation:** Online Webex. A date within 6 - 26 July 2025. Webex link will be provided later.

Good Luck! #DesignLikeAPro